



TB AUDIO PLUG-INS
Team Impulse Impact

USER MANUAL

TB Limiter

True Peak Mastering Limiter

A stereo-linked true-peak limiter with a fixed 2 ms lookahead, program-aware release, micro-peak Enhance, and optional TPDF dither.

Ceiling -3 to 0 dBTP

Fixed 2 ms lookahead

8 controls

9 factory presets



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01 Overview

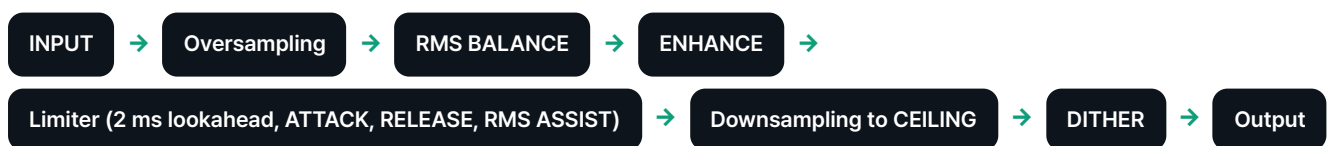
TB Limiter is a stereo-linked true-peak mastering limiter built around eight decisive controls. It is not a sample clipper: **CEILING** is a true-peak bound, checked with fixed high-quality oversampling.

Drive density with **INPUT**, set an independent final **CEILING**, shape micro-peaks with **ENHANCE**, and control recovery with **RELEASE** and the program-aware **RMS ASSIST**. **RMS BALANCE** gently corrects persistent left-right energy drift, and **ATTACK** decides how the fixed 2 ms lookahead approaches each coming peak. Optional 16-bit or 24-bit TPDF dither completes the final delivery stage.

Key features

- True-peak ceiling from -3 to 0 dBTP
- Stereo-linked limiting with a fixed 2 ms lookahead and fixed oversampling
- Release with program-aware RMS Assist: faster for isolated peaks, slower for dense material
- Enhance for extra density by softening and driving short peaks
- RMS Balance for slow left-right correction
- Optional 16-bit or 24-bit TPDF dither
- Gain reduction, true-peak, and left-right balance displays
- Works from 44.1 to 192 kHz with latency-aligned bypass

Signal flow



About this manual

This manual describes TB Limiter 1.2.0 for Windows (VST3). The current macOS build is version 0.4.2 and may differ from the descriptions here.

02 Quick start

- 1 Insert TB Limiter last on the master bus. The defaults are **INPUT** 4 dB, **CEILING** -1 dBTP, **RELEASE** 120 ms, **RMS ASSIST** 65%, and **ATTACK** 2 ms.

- 2 Set **CEILING** 5 to the true-peak limit of your delivery. -1 dBTP is a common choice for streaming.
- 3 Raise **INPUT** until the music is as loud and dense as you want, watching LIMIT GR 2.
- 4 Adjust **RELEASE** and **ATTACK** 7, and balance recovery with **RMS ASSIST** 8.
- 5 Add **ENHANCE** 6 only if you need more density.
- 6 Turn on **DITHER** 9 only for the final 16-bit or 24-bit export, and only if your DAW does not dither.

03 Interface



TB Limiter interface.

- 1 **Preset bar** — previous and next preset, the preset list, Save, Undo, and Redo.
- 2 **PEAK GOVERNOR** — LIMIT GR shows the gain reduction on a 0 to 12 dB scale.
- 3 **TRUE PEAK** — the output true-peak level, with the CEILING marker.
- 4 **L/R BALANCE** — the current left-right balance correction.
- 5 **INPUT, CEILING** — drive and the true-peak limit.
- 6 **ENHANCE** — micro-peak shaping before the limiter.
- 7 **ATTACK, RELEASE** — how gain approaches peaks and recovers.
- 8 **RMS ASSIST, RMS BALANCE** — program-aware recovery and left-right correction.
- 9 **DITHER** — Off, 16 BIT, or 24 BIT.

04 Controls

Control	Range · default	What it does
INPUT	0 to 18 dB · 4 dB	Sets how hard the signal drives the limiter. More input means more gain reduction and density; the ceiling stays the same, so compare at matched loudness.
CEILING	-3 to 0 dBTP · -1 dBTP	Sets the maximum true-peak output level.
ENHANCE	0 to 100% · 0%	Adds density by softening and driving short peaks before the limiter. The upper half is intentionally aggressive and can change colour.
ATTACK	0.1 to 2 ms · 2 ms	Sets how quickly gain moves toward a predicted peak within the fixed 2 ms lookahead. It does not change the latency.

Control	Range · default	What it does
RELEASE	20 to 500 ms · 120 ms	Sets how quickly limiting recovers. Very short values can pump.
RMS ASSIST	0 to 100% · 65%	Adapts recovery: faster for isolated peaks and slower for dense material. It blends with RELEASE rather than replacing it.
RMS BALANCE	0 to 100% · 0%	Gently corrects persistent left-right level imbalance. It reacts slowly and does not correct short stereo movement.
DITHER	Off, 16 BIT, 24 BIT · Off	Adds final TPDF dither for 16-bit or 24-bit delivery. Leave it off when the DAW will dither.

Latency and dither

TB Limiter has a fixed latency: the 2 ms lookahead plus the oversampling filters. The plug-in reports it to the host and keeps bypass latency-aligned, so the track stays in time. TB Limiter has no compressor stage, no multiband or spectral processing, no separate clipper, and no automatic loudness target.

Dither only adds TPDF noise; the host or exporter still converts to the final word length. Use it only on the very last export, after all gain changes and sample-rate conversion, and never on float files, stems, or when the DAW already dithers.

05 Presets

Step through presets with the arrows, or open the list to choose a factory or user preset. Save stores the current settings as a user preset. Undo and Redo step through recent parameter changes. All factory presets use a -1 dBTP ceiling and a 2 ms attack, with dither off.

Preset	INPUT	RELEASE	ENHANCE	RMS ASSIST	RMS BALANCE
Init	4 dB	120 ms	0%	65%	0%
Transparent	2 dB	180 ms	0%	40%	0%
Punch	5 dB	70 ms	0%	80%	0%
Dense Control	6 dB	250 ms	0%	75%	0%
Balance Guard	4 dB	150 ms	0%	65%	100%
Maximize	8 dB	180 ms	55%	75%	0%
Simple Limiting	0 dB	120 ms	0%	0%	0%
Mastering	4 dB	160 ms	30%	65%	0%
RMS Smooth	3 dB	180 ms	0%	100%	50%

06 Recipes

Starting points for common jobs. Adjust them by ear.

Streaming master

A clean, competitive master with true-peak headroom.

Preset	Mastering
CEILING	−1 dBTP
INPUT	Until LIMIT GR peaks at 2 to 4 dB

- 1 Compare with bypass at matched loudness.
- 2 Check the loudest section, not just the start.

Transparent peak control

Catch peaks with as little colour as possible.

Preset	Transparent
INPUT	0 to 3 dB
ENHANCE	0%

- 1 Lengthen RELEASE if you hear pumping.

Dense, loud mix

Electronic or rock material that should feel powerful.

Preset	Dense Control or Maximize
ENHANCE	20 to 55%

- 1 Above 50%, ENHANCE becomes aggressive; listen for harshness.

Final 16-bit export

A CD-format or other 16-bit file.

DITHER	16 BIT
DAW dither	Off

- 1 Turn dither on only for the final export.

07 Tips and troubleshooting

Symptom	What to check
The master pumps.	Lengthen RELEASE or raise RMS ASSIST, and reduce INPUT.
Transients sound dull.	Use less INPUT, a slightly shorter ATTACK, or less ENHANCE.
It distorts at high ENHANCE.	The upper half of ENHANCE is intentionally aggressive. Lower it.
True peaks exceed the ceiling after export.	Later gain changes or sample-rate conversion can raise peaks. Put TB Limiter last and verify the exported file.
Noise appears in quiet parts.	DITHER is on. Turn it off unless this is the final 16-bit or 24-bit export.
The left-right balance shifts slowly.	RMS BALANCE is correcting a persistent imbalance. Lower it if that is not wanted.

08 Parameter reference

Panel label	Host parameter	Range	Default
INPUT	Input Gain	0 to 18 dB	4 dB
CEILING	Ceiling	−3 to 0 dBTP	−1 dBTP

Panel label	Host parameter	Range	Default
RELEASE	Release	20 to 500 ms	120 ms
RMS ASSIST	RMS Assist	0 to 100%	65%
RMS BALANCE	RMS Balance	0 to 100%	0%
ENHANCE	Enhance	0 to 100%	0%
ATTACK	Attack	0.1 to 2 ms	2 ms
DITHER	Dither	Off, 16-bit, 24-bit	Off

09 Version history

Version	Changes
1.2.0 · 2026-08-02	Applied a faster mapping across the 50–100% ENHANCE range with up to 3 dB of pre-limiter drive, and adjusted the true-peak guard at high ENHANCE settings. Parameters and saved-state compatibility are unchanged.

10 Support and license

Report problems or request features at github.com/TaebeumKim/TB_Audio-Plugins/issues. Include the plug-in version, your DAW, and your operating system.

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